

COMMUNICATION SYSTEMS SIMULATION FRAMEWORK

Comprehensive Analytical & Mathematical Specifications Report

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Abstract: This documentation compiles an exhaustive mathematical, algorithmic, and programmatic framework of the end-to-end communication systems simulator. The platform models discrete-time processing across signal synthesis, complex passband analog modulations (including standard AM, coherent DSB-SC, analytic Hilbert-based SSB, and integrated FM), physical medium modeling via linear time-invariant channel convolution kernels, non-coherent and coherent receiver architectures, and uniform Pulse Code Modulation (PCM) digitalization.

1. BASEBAND INFORMATION SOURCE MODEL (`INFOSOURCE.PY`)

The information source synthesizes continuous baseband message waveforms denoted as $m(t)$. The simulator supports both deterministic test tones and randomized stochastic representations:

- **Deterministic Single-Tone Harmonic:** Used to assess baseline phase and harmonic response parameters:

$$m(t) = A_m \sin(2\pi f_m t)$$

Where A_m represents the baseband peak amplitude scale and f_m specifies the fundamental cyclic frequency.

- **Deterministic Multi-Tone Signal:** Simulates broader spectral voice/data structures without exceeding peak-to-peak constraints by introducing a normalization scalar K :

$$m(t) = \frac{1}{K} \sum_{k=1}^K A_{m_k} \sin(2\pi f_k t)$$

Where f_k is an array of independent constituent spectrum frequencies.

- **Stochastic Normal White Noise Process:** Simulates unstructured baseline noise arrays:

$$m[n] \sim \mathcal{N}(\mu, \sigma^2)$$

2. MATHEMATICAL FRAMEWORK OF TRANSMIT MODULATORS (`TXMOD.PY`)

Transmitter modulation converts baseband profiles into high-frequency passband channels mapped against a harmonic carrier reference $c(t) = A_c \cos(2\pi f_c t)$, where $f_c \gg f_m$.

2.1 Double Sideband Suppressed Carrier (DSB-SC)

DSB-SC eliminates carrier power dissipation by directly multiplying the message and carrier vectors, resulting in a phase inversion when the message crosses zero:

$$x_{DSB}(t) = m(t) \cdot \cos(2\pi f_c t)$$

2.2 Standard Amplitude Modulation (AM)

Standard AM introduces a deterministic DC bias offset factor A_{dc} to ensure that the message envelope remains strictly non-negative. This design choice enables the use of simple, low-cost envelope detectors at the receiver:

$$x_{AM}(t) = [m(t) + A_{dc}] \cdot \cos(2\pi f_c t)$$

To avoid overmodulation distortion, the modulation index μ_{AM} must satisfy the envelope stability condition:

$$\mu_{AM} = \frac{\min(|m(t)|)}{A_{dc}} \leq 1$$

2.3 Single Sideband Modulation (SSB) via Analytic Hilbert Transformers

SSB modulation maximizes spectral efficiency by suppressing one of the redundant mirror sidebands. This framework uses a 90-degree phase shifter implemented via the continuous-time Hilbert Transform integral:

$$\hat{m}(t) = \mathcal{H}\{m(t)\} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{m(\tau)}{t - \tau} d\tau$$

In the frequency domain, this transform applies an asymmetric phase shift operator:

$$H(f) = -j \cdot \text{sgn}(f)$$

The time-domain passband signals for Upper Single Sideband (USSB) and Lower Single Sideband (LSSB) are synthesized using the following quadrature expressions:

$$x_{SSB}(t) = m(t) \cos(2\pi f_c t) \mp \hat{m}(t) \sin(2\pi f_c t)$$

Where the subtraction operator (-) isolates the Upper Sideband (x_{USSB}) and the addition operator (+) isolates the Lower Sideband (x_{LSSB}).

2.4 Frequency Modulation (FM)

FM is a non-linear constant-envelope modulation technique where the instantaneous carrier frequency varies linearly with the baseband message amplitude. The continuous-time expression is given by:

$$x_{FM}(t) = A_c \cos \left(2\pi f_c t + 2\pi k_f \int_{-\infty}^t m(\tau) d\tau \right)$$

Where k_f represents the frequency sensitivity factor. In a discrete-time processing framework, the integration is evaluated using numerical rectangular summation via cumulative accumulation (`np.cumsum`):

$$\theta[n] = 2\pi f_c t[n] + 2\pi k_f \cdot \Delta t \sum_{i=0}^n m[i]$$

Where the discrete time step is defined by the sampling interval $\Delta t = 1 / f_s$.

3. PHYSICAL CHANNEL MODELING AND IMPAIRMENTS (`CHANNEL.PY`)

The physical medium is modeled as a Linear Time-Invariant (LTI) system where the received signal is the convolution of the transmitted signal and the channel's impulse response: $r(t) = x(t) * h(t)$.

- **Distortionless Attenuation & Propagation Delay Channel:** Simulates an ideal line-of-sight path with flat frequency response:

$$h_{dist}(t) = a \cdot \delta(t - t_d) \implies r(t) = a \cdot x(t - t_d)$$

Where a represents the channel attenuation factor and t_d specifies the physical propagation time delay.

- **Bandlimited Brick-Wall Channel:** Simulates an ideal lowpass filter channel that completely truncates high-frequency spectral components above a strict cutoff bandwidth B :

$$h_{band}(t) = 2 a B \cdot \text{sinc}(2 B t) = 2 a B \frac{\sin(2\pi B t)}{2\pi B t}$$

4. RECEIVER DEMODULATION TOPOLOGIES (`RX_DEMOD.PY`)

The receiver subsystem processes the degraded channel output to recover a high-fidelity estimate of the original baseband signal, denoted as $\hat{m}(t)$.

4.1 Coherent Synchronous Demodulation (SD)

Synchronous demodulation requires a local oscillator that is perfectly phase-locked with the transmitter's carrier reference. Multiplying the received DSB-SC signal by this local carrier shifts the baseband content back to baseband while generating a double-frequency image component:

$$y(t) = x_{DSB}(t) \cdot \cos(2\pi f_c t) = m(t) \cos^2(2\pi f_c t) = \frac{1}{2}m(t) + \frac{1}{2}m(t)\cos(4\pi f_c t)$$

An ideal windowed sinc lowpass filter kernel (`filter_sinc.py`) is convolved with the signal (`convo_out.py`) to eliminate the high-frequency term centered at $2f_c$:

$$\hat{m}(t) = y(t) * g_{sinc}(t) = \frac{1}{2}m(t)$$

4.2 Non-Coherent Envelope Detection (ED) via Analytic Signals

Envelope detection simplifies receiver design by tracking the absolute amplitude variations of the received signal. In the simulator, this is implemented using an analytic complex representation via the Hilbert Transform, which tracks the instantaneous envelope without requiring a coherent local carrier:

$$z(t) = x_{AM}(t) + j \mathcal{H}\{x_{AM}(t)\}$$

$$\hat{m}(t) = |z(t)| - A_{dc} = \sqrt{x_{AM}^2(t) + \hat{x}_{AM}^2(t)} - A_{dc}$$

4.3 FM Discriminator Demodulation (EDFM) via Analytic Phase Unwrapping

FM demodulation extracts the message signal from the derivative of the instantaneous phase. The receiver maps the passband signal into its complex analytic form to extract the phase angle profile:

$$\phi[n] = \text{unwrap}\left(\text{atan}^{-1}\left(\frac{\mathcal{H}\{x_{FM}[n]\}}{x_{FM}[n]}\right)\right)$$

The message signal is proportional to the time derivative of this phase profile. This derivative is calculated numerically using a discrete backward-difference operator, followed by a mean correction step to remove any residual DC component:

$$\hat{m}[n] = \frac{1}{2\pi k_f} \left(\phi[n] - \phi[n-1] \right) / \Delta t - \mu_{\hat{m}}$$

5. DIGITAL SIGNAL DIGITALIZATION BLOCK (`ADC.PY`)

The Analog-to-Digital Converter (ADC) transforms continuous-amplitude baseband signals into discrete binary sequences using Pulse Code Modulation (PCM). This process consists of two primary stages: uniform quantization and binary encoding.

- **Uniform Quantization Step Size Resolution (Δ_q):** Given an allocation of N bits, the input dynamic range is divided into $L = 2^N$ discrete levels over the interval $[m_{min}, m_{max}]$:

$$\Delta_q = \frac{m_{max} - m_{min}}{2^N - 1}$$

- **Quantization Index Encoder Assignment Mapping:** Each continuous sample is mapped to its nearest valid quantization index using a round-and-clip operation:

$$I[n] = \text{clip}\left(\left\lfloor \frac{m[n] - m_{min}}{\Delta_q} + 0.5 \right\rfloor, 0, 2^N - 1\right)$$

These indices $I[n]$ are subsequently converted into standard N -bit binary strings.

6. FRAMEWORK MODULE CROSS-REFERENCE INDEX

Module Filename	Primary Algorithmic Target	Core Mathematical Variables
<code>infosource.py</code>	Baseband Signal Synthesis & Waveform Audio Ingestion	$A_m, f_m, K, \mathcal{N}(\mu, \sigma^2)$
<code>txmod.py</code>	Passband Analog Modulation Engines (AM, DSB, SSB, FM)	$A_c, f_c, k_f \text{heta}[n], \mathcal{H}\{m(t)\}$
<code>channel.py</code>	LTI Media Attenuation, Delay & Sinc Lowpass Truncation	$a, t_p, B, \delta(t), \text{ext}\{\text{sinc}\}(x)$
<code>rx_demod.py</code>	Coherent Sync Trackers, Analytic Envelopes & Phase Differentiators	$g_{\text{sinc}}(t), z(t) , \text{ext}\{\text{unwrap}\}(\phi)$
<code>adc.py</code>	Uniform Quantization Linear Arrays & Binary PCM Bitstream Generation	$N, 2^N, \Delta_q, I[n]$
<code>filter_sinc.py</code>	Discrete Windowed Sinc Filter Impulse Response Generation	$B, f_s, g[n] = 2B \cdot \text{ext}\{\text{sinc}\}(2B n \Delta t)$

7. SUMMARY OF KEY SIMULATION VARIABLES

Variable Symbol	Technical Definition	Dimensional Unit Context
$m(t) / m[n]$	Continuous / discrete baseband target input information source message	Volts [V] / Dimensionless Normalized Array
$x(t) / r(t)$	Modulated passband transmitted wave / Physical corrupted media output	Volts [V] Baseline Potential Difference
$f_m / f_c / f_s$	Message frequency / Passband carrier center / Sampling processing frequency	Hertz [Hz] (Cycles per Second)
k_f	Frequency deviation sensitivity parameter of the voltage-controlled oscillator	Hertz per Volt [Hz/V]
$\hat{m}(t)$	Hilbert Transform quadrature projection phase-shifted image element	Volts [V] orthogonal domain trace
$\hat{m}(t) / \hat{m}[n]$	Reconstructed baseband output estimate evaluated at the receiver stage	Volts [V] / Rebuilt numerical vector